A Preference-Optimization Comparison Where the Reward Is Exact, and Where the Statistics Almost Hid the Result

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Pairwise vs. Group-Relative Optimization Under a Programmatic Verifier | September 2026

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Abstract

Comparing preference-optimization methods normally means comparing them under a learned, noisy reward model, which confounds the optimization method with reward-model error. We remove that confound by using an exact programmatic verifier as the reward and comparing pairwise (DPO) against group-relative (GRPO) optimization from an identical supervised-fine-tuning checkpoint, across ten seeds. Both methods roughly triple held-out accuracy over the policy they start from ($+0.180$ and $+0.191$, each exceeding 6σ of the baseline's own seed spread). The comparison between the two methods themselves is where the more interesting result lies: the standard unpaired test calls the gap statistically indistinguishable from noise (0.25σ), while the paired test the experimental design actually supports — both methods share an SFT checkpoint per seed, so per-seed quality is a nuisance term that pairing cancels — finds one method ahead on all ten seeds ($p = 0.00098$). The same data, two tests, two conclusions; only one respects the design. A follow-up sweep intended to explain the effect mechanistically returns an inconclusive, non-monotonic pattern, which we report as such rather than reshaping into a story the data does not clearly tell.

1 Motivation

Comparing two preference-optimization algorithms under a learned reward model conflates two sources of variation: differences in how the algorithms use preference signal, and noise or bias in the reward model that produced that signal. A programmatic, exact verifier removes the second source entirely — a completion either satisfies the verifier or it does not, deterministically — so any measured difference between algorithms is attributable to the algorithms themselves.

2 Setup

Task. Two-digit addition with a fixed-width, zero-padded three-digit answer, so every completion has identical token length and the reward carries no length confound. The verifier is a Python function: exact match on the correct sum, zero otherwise.

Policy. A from-scratch decoder-only Transformer (151,872 parameters at this task's vocabulary size), with no pretrained initialization.

Pipeline. Supervised fine-tuning (SFT) on correct (prompt, answer) pairs, deliberately stopped short of solving the task so that headroom remains for preference optimization to act on, followed by *both* DPO and GRPO branching from the identical SFT checkpoint per seed — branching both methods from one checkpoint, rather than giving each its own SFT run, removes per-run SFT quality as a confound between methods.

DPO and GRPO are implemented behind a single method flag sharing one training loop, one rollout procedure, one evaluator, and one frozen reference model, so that only the loss function differs between them:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma(\beta\,[(\log p_\theta(y_w) - \log p_{\text{ref}}(y_w)) - (\log p_\theta(y_l) - \log p_{\text{ref}}(y_l))])$$

$$\mathcal{L}_{\text{GRPO}} = -\,\mathbb{E}[A_i \log p_\theta(y_i)] + \beta\,\widehat{\text{KL}}(p_\theta \,\|\, p_{\text{ref}}), \qquad A_i = \frac{r_i - \bar{r}_{\text{group}}}{\text{std}(r_{\text{group}})}$$

DPO reduces each sampled group to its best/worst pair; GRPO uses the full group of K samples via a group-normalized advantage in place of a learned value baseline.

3 Correctness Verification

Two checks were run before trusting any downstream number. First, the per-sequence completion log-probability routine — the primitive both losses depend on, and the most common source of a silently-wrong preference-optimization implementation — was checked against a brute-force, token-by-token recomputation: measured discrepancy 0.0 across a test batch. Second, at initialization the policy equals the reference model by construction, so the DPO loss must equal exactly $\log 2 = 0.6931$; this was confirmed to the observed precision on every seed's first training step.

4 Result: Preference Optimization Triples Accuracy

	Held-out accuracy	vs. SFT
SFT	0.108 ± 0.050	—
DPO	0.288 ± 0.046	$+0.180$ (3.6σ)
GRPO	0.299 ± 0.047	$+0.191$ (3.9σ)

Table 1: 10 seeds, group size $K = 8$. Both methods clear 3.5σ against the SFT policy's own seed spread.

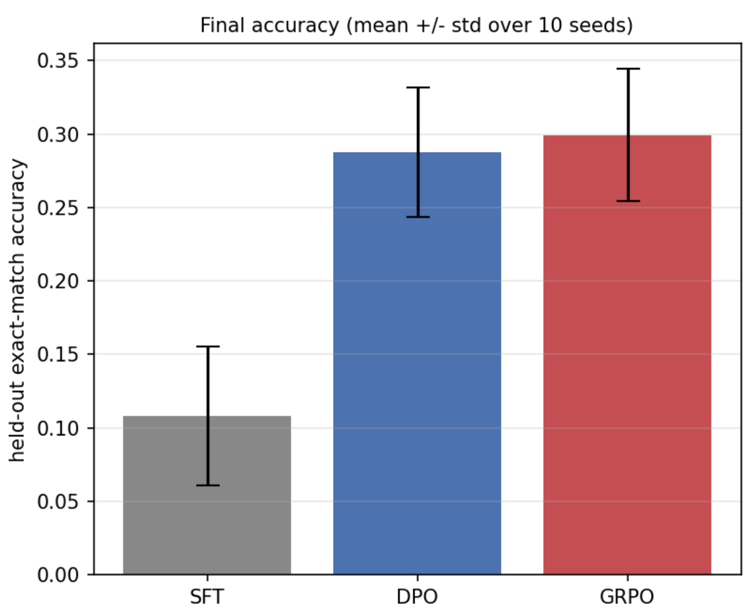

Figure 1: Held-out accuracy, mean ± std over 10 seeds. DPO and GRPO's error bars overlap heavily — the reason the unpaired test in Section 5 misses the effect the paired test finds.

A group-composition diagnostic explains why: the fraction of sampled groups containing both a correct and an incorrect completion — the only groups either method can learn from — rises from 0.38 to 0.81 over training on a representative seed, confirming both methods depend on, and increasingly exploit, partial policy correctness as training progresses.

A tempting further observation is that final accuracy correlates strongly with how weak the SFT checkpoint was: $r = -0.871$ between SFT accuracy and GRPO's *gain* over it. That number should not be trusted as stated — gain is computed as final minus SFT, so SFT appears on both sides with opposite sign, and a merely noisy SFT measurement manufactures exactly this kind of correlation with no underlying effect. The clean version, SFT accuracy against *final* accuracy directly, where no shared term exists, gives $r = -0.491$ ($t = -1.60$, $df = 8$): the sign survives, the significance does not at $n = 10$. This is reported as a hint, not a finding.

5 DPO vs. GRPO: The Test That Is Used Matters

Read as independent samples, DPO and GRPO are indistinguishable: a gap of $+0.0117$ against DPO's own standard deviation of 0.046 is 0.25σ. But both methods share an SFT checkpoint on every seed, so per-seed SFT quality is a nuisance term common to both — a seed whose SFT run happened to land high lifts both methods together, and an unpaired comparison throws that shared structure away.

Seed	42	43	44	45	46	47	48	49	50	51
GRPO−DPO	+.014	+.012	+.004	+.012	+.018	+.016	+.023	+.004	+.012	+.004

Table 2: Paired difference, every one of ten seeds. Mean $+0.0117 \pm 0.0064$; paired $t = 5.75$ ($df = 9$); GRPO ahead on 10/10 seeds, sign-test $p = 0.00098$.

The same ten seeds support two opposite readings depending only on whether the comparison respects the paired structure the experiment was designed with; the unpaired test is not merely less powerful here, it is the wrong test for this design.

The effect itself is small in absolute terms (+1.2 percentage points) and is reported at that size rather than inflated; what makes it a result rather than noise is its consistency across every independently-seeded run, not its magnitude.

6 An Attempted Mechanism Test

GRPO uses all K sampled completions per prompt; DPO reduces each group to one pair. If that sample-efficiency difference is what produces Table 2's effect, the gap should grow with K and vanish at $K = 2$, where both methods see identical data. Sweeping $K \in \{2, 4, 8, 16\}$ at five seeds each does not show that pattern (Table 3, Figure 2).

K	DPO	GRPO	Paired Δ (SE)	Seeds ahead
2	0.268	0.267	−0.001 (0.007)	3/5
4	0.294	0.285	−0.009 (0.005)	1/5
8	0.306	0.318	+0.012 (0.002)	5/5
16	0.321	0.307	−0.014 (0.008)	1/5

Table 3: No monotonic trend: the gap spikes at $K = 8$ — the value used throughout Sections 4–5 — and is not distinguishable from zero, or favors DPO, at every other K tested.

Five seeds per K is underpowered to distinguish a genuine $K = 8$-specific effect from noise at this

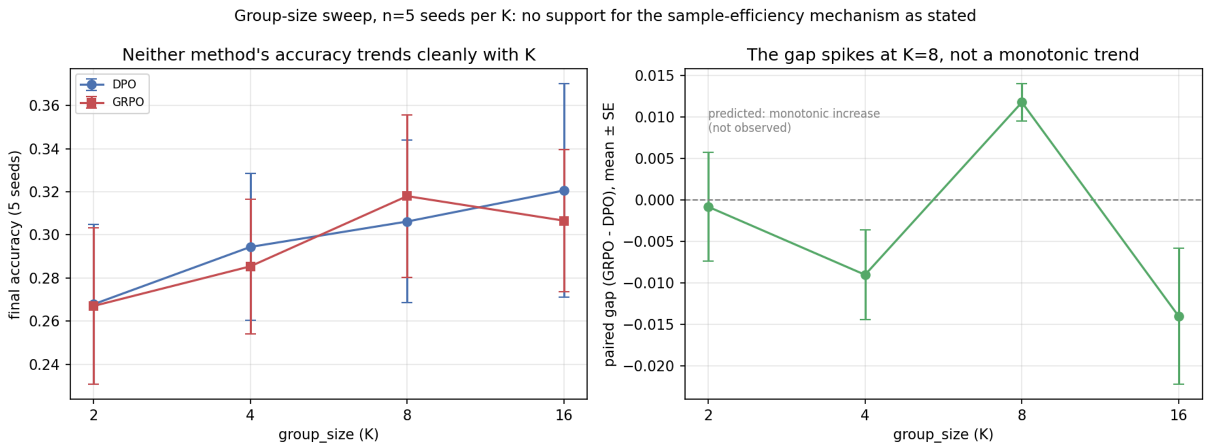

Figure 2: Left: neither method's accuracy trends cleanly with K. Right: the paired gap by K — a spike, not the predicted monotonic increase.

sample size — and notably smaller than the ten seeds the headline result in Section 5 required before its effect was reliably detectable — so the honest reading is that this experiment neither confirms nor refutes the sample-efficiency hypothesis. One incidental result is unambiguous: the $K = 8$ arm, rerun independently at five of the ten original seeds, reproduces Table 2's per-seed differences to within 0.001 on every seed, confirming the pipeline is deterministic enough for a fixed seed to mean what it claims to mean.

7 Discussion

The headline methodological point is general, not specific to DPO or GRPO: whenever a comparison shares a nuisance variable across the objects being compared — here, a common SFT starting point — an unpaired test discards information the experimental design already provides for free, and can report noise where a paired test reports a small, consistent, real effect. The group-size sweep is a caution against the opposite error: having found an effect, the natural next step of explaining it mechanistically is not guaranteed to succeed even when the headline effect is solid, and an inconclusive follow-up should be reported as inconclusive rather than narrated into a mechanism the data does not clearly support.

8 Limitations

- Two-digit addition with an exact verifier is a controlled setting, not a benchmark; generalization to open-ended tasks with learned or partial-credit reward is untested.
- The correlation between SFT quality and the size of the preference-optimization gain is confounded by shared terms when computed as SFT-vs-gain; a cleaner SFT-vs-final-accuracy test shows the same sign but does not reach significance at $n = 10$.
- The group-size sweep (Section 6) is underpowered by design; it rules out neither a real K-dependent effect nor a purely spurious $K = 8$ result.
- Policy and reward model share no pretrained initialization; whether the paired effect in Table 2 persists with a pretrained base model is untested.

9 Path Forward

The clearest next step is more seeds on the group-size sweep — ten per K, matching Section 5's design, would likely settle whether $K = 8$ is genuinely special or simply the point where five seeds happened to land favorably. A second, complementary experiment varies SFT checkpoint quality directly (via SFT step count) rather than leaving it to chance across seeds, converting Section 4's incidental correlation into a controlled test of whether the two methods degrade differently as the base policy approaches 0% or 100% success.

10 Reproducibility

Code, both methods, the correctness checks, all ten seeds, and the group-size sweep: `https://github.com/saqarkpr/verifier-based-preference-optimization`.